

TODO: Short Paper Title

Anonymous ACL submission

Abstract

Chemical knowledge is produced, searched, and reused across language boundaries: a technical question may be asked in one language while the strongest evidence appears in another. Yet multilingual chemistry retrieval has little clean public evaluation data, so models are often judged by aggregate recall even when that number hides the failure that matters most: retrieving the right evidence across languages rather than an easier same-language match. We address this gap with a multilingual chemistry QAC benchmark and generation pipeline for cross-lingual retrieval evaluation. Patents provide the benchmark substrate in this work because multilingual publication variants and chemistry ontologies make it possible to construct controlled relevance judgments and chemically plausible hard negatives. The pipeline extracts chemistry-relevant technical text, generates multiple question-answer-context candidates with an LLM, grades them for faithfulness and retrieval quality, and validates the automated grading against human annotation. We export the resulting data in retrieval-ready form with corpus, queries, qrels, and QAC records, enabling evaluation by query language, document language, same-language versus cross-language relevance, and chemically confusable hard negatives. Across evaluated multilingual embedding models, average performance masks systematic cross-language weakness: models can prefer same-language or chemically similar but irrelevant documents over the intended evidence. The benchmark and pipeline provide a practical framework for building and auditing specialized multilingual retrieval evaluations in high-trust technical domains. Data and code are available on [Hugging Face](#) and [GitHub](#).

1 Introduction

Technical search in chemistry rarely respects a single language boundary. A researcher may ask a question in English, German, French, Spanish, or

another working language, while the strongest evidence may be documented elsewhere: in a translated patent family, in a regional filing, or in technical text whose terminology follows a different linguistic convention. This is not only a translation problem. Chemistry search depends on aliases, abbreviations, formulae, compound families, process descriptions, and near-neighbor concepts that can be semantically close while still being wrong for the user’s question. For a retrieval system used in a high-trust technical workflow, returning a plausible but chemically different document is not a harmless ranking error.

Dense retrieval with neural sentence and passage embeddings (Reimers and Gurevych, 2019; Karpukhin et al., 2020) is attractive for this setting because multilingual embedding models promise a single retrieval index over many languages (Feng et al., 2022; Chen et al., 2024). In practice, however, the evaluation signal used to choose such a model is often too coarse. A model can achieve a strong average Recall@k by retrieving easy same-language evidence, while failing when the intended evidence is in another language, a bias also observed in recent cross-lingual retrieval analyses (Goworek et al., 2025; Amiraz et al., 2025). The problem is not only whether a multilingual encoder retrieves a relevant document, but whether it retrieves the right document when query language, document language, and chemical similarity pull the ranking in different directions. This is the failure mode that matters in deployment, and it is precisely the one that an aggregate retrieval score can hide.

Existing multilingual retrieval benchmarks provide important coverage for general cross-lingual search (Zhang et al., 2023; Lawrie et al., 2023; Muennighoff et al., 2023; Enevoldsen et al., 2025), but they do not directly capture this chemistry workflow. We need evaluation data where language variants are controlled, questions are

grounded in technical contexts, and hard negatives are chemically plausible rather than arbitrary. Without this structure, an evaluation may reward the model that looks best on average while missing the model behavior that creates practical risk.

We present a multilingual chemistry question-answer-context (QAC) benchmark and generation pipeline for cross-lingual retrieval evaluation. The broader task is multilingual chemistry retrieval; patents are the substrate we use to construct the benchmark because they provide multilingual technical variants and chemistry-rich descriptions at scale, building on the long history of patent retrieval evaluation (Piroi et al., 2011). The pipeline generates and grades QAC records, validates automated grading against human annotation, and exports corpus, queries, qrels, and QAC tables for standard retrieval evaluation.

The benchmark is designed to make hidden retrieval failures visible. It separates same-language and cross-language relevance, supports analysis by query and document language, and includes chemistry-aware hard negatives derived from alias and ontology structure, including ChEBI (Hastings et al., 2016). This lets us ask whether a model retrieves the intended cross-language evidence instead of a same-language shortcut or a chemically similar distractor.

This paper is therefore an evaluation-framework paper rather than a model leaderboard. The contribution is a practical way to build, validate, and audit specialized multilingual retrieval benchmarks in a domain where both language and technical similarity matter. Our contributions are a public multilingual chemistry QAC benchmark, a reproducible LLM-assisted generation and validation pipeline, a language-aware retrieval evaluation setup, and a chemistry-confusability stress test based on alias and ontology structure.

Together, these components provide a reusable lesson for high-trust technical retrieval: start from the workflow failure, build data that makes the failure observable, and report language-aware robustness alongside standard retrieval metrics.

2 Related Work

Multilingual retrieval benchmarks such as MIRACL, NeuCLIR, MTEB, MMTEB, and CLIRMatrix have made cross-lingual retrieval evaluation easier to compare across models and languages (Zhang et al., 2023; Lawrie et al., 2023; Muen-

nighoff et al., 2023; Enevoldsen et al., 2025; Sun and Duh, 2020). They are closest to our evaluation setting, but they are not designed around chemistry-specific relevance, controlled language variants, or chemically plausible hard negatives. Our benchmark follows this line of work while adding domain structure needed to audit multilingual retrieval in technical chemistry search.

Recent work on cross-lingual retrieval and multilingual RAG shows that model rankings can depend strongly on language, alignment, and same-language preference (Goworek et al., 2025; Amiraz et al., 2025; Li et al., 2024; Liu et al., 2025; Ki et al., 2025). These findings motivate our focus on separating same-language from cross-language relevance instead of relying only on aggregate recall. In contrast to settings where content and language are difficult to disentangle, patent publication variants give us a practical way to compare multilingual technical evidence under tighter content control.

Patent retrieval has a long evaluation history, including NTCIR and CLEF-IP (Fujii and Ishikawa, 2002; Piroi et al., 2011), and recent work has revisited patent embeddings (Ghosh et al., 2024; Yousefiramandi and Cooney, 2026). Our goal is complementary: we do not propose a new encoder, but a chemistry-grounded evaluation framework. We use chemical ontology and alias structure, including ChEBI (Hastings et al., 2016), to define hard negatives that are plausible for chemistry search but wrong for the intended evidence.

3 QAC Generation Pipeline

Our method turns multilingual chemistry technical text into auditable question-answer-context (QAC) records for retrieval evaluation. The design goal is not simply to generate fluent questions. Each generated item must identify the evidence document it came from, preserve language metadata, and carry quality signals that make the benchmark inspectable after export. Figure 1 summarizes the full pipeline.

Corpus and context construction. The input is a chemistry-relevant multilingual corpus. In this work, we use patent-derived text as the benchmark substrate because patents provide technical descriptions and multilingual publication variants at scale. For each document-language pair, the context builder assembles a compact evidence block from available fields such as title, abstract, and

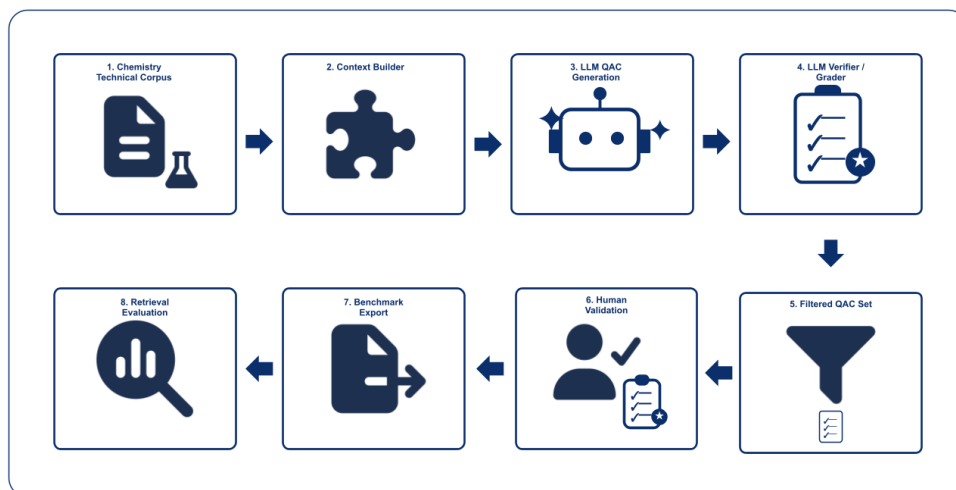


Figure 1: Overview of the multilingual chemistry retrieval benchmark pipeline, from chemistry technical text through QAC generation, verification, human validation, benchmark export, and retrieval evaluation.

Figure note: Icons adapted from Font Awesome Free, licensed under CC BY 4.0.

technical description or claim text. This step keeps the generation grounded in a specific document rather than asking the model to produce generic chemistry questions from memory.

Candidate generation. For each selected document, language, and generation mode, an LLM produces multiple QAC candidates. Each candidate contains a user-facing question, a short answer, the supporting context, and metadata linking the item back to the source document and language. We use both technical and semantic question modes so that the benchmark contains queries that reflect different retrieval intents: some ask for concrete technical details, while others ask for a higher-level description of what the document is about.

Verification and filtering. Generated candidates are then passed through an LLM-based verifier. The verifier scores whether the answer is faithful to the provided context, whether the question is answerable from that context, whether the language is appropriate, and whether the item is useful as a retrieval query. We retain the best-scored items together with their audit metadata rather than treating generation as a black box. This gives each QAC row a provenance trail: source document, language, generation mode, scores, and verifier feedback.

Human validation and export. Because the verifier is itself model-based, we validate its be-

havior against a human-annotated sample. This does not replace expert review for every item, but it checks whether the automatic grading is directionally reliable enough for benchmark construction. The filtered and validated QAC records are then exported with the retrieval artifacts used in later sections: a corpus table, query table, relevance judgments, and the full QAC table. These artifacts let us evaluate models with standard dense-retrieval tooling while still slicing results by query language, document language, and source provenance.

4 Benchmark Construction

The filtered QAC records are exported as retrieval benchmarks rather than as a single flat table. Each release contains four linked views: corpus stores the searchable documents and their languages; queries stores the generated questions; qrels stores relevance judgments from queries to documents; and qac preserves the original question, answer, context, source document, generation metadata, and quality scores. This separation lets standard retrieval evaluators consume the data while keeping the provenance needed to audit failures by language and source.

Cross-language relevance. For the multilingual chemistry retrieval benchmark, each query is tied to a source technical document. Relevant documents include the source document and its available multilingual variants, so a query can be

evaluated against both same-language and cross-language evidence. We keep query language, document language, and source identifiers as explicit metadata. This makes it possible to separate an easy same-language hit from a harder cross-language hit, and to construct “no-home” cases in which the query language has no same-language gold document. These cases are central to the benchmark because they remove the shortcut that often inflates aggregate recall.

Chemistry-aware hard negatives. Language is only one source of retrieval error in chemistry. A model may also rank a document about a related compound, synonym, parent class, or neighboring concept above the intended evidence. To test this failure mode, we build an alias-graph stress test from chemistry ontology and alias structure. ChEBI concepts and multilingual aliases define target chemical concepts; documents associated with the target concept form the gold set; and taxonomic or alias-graph neighbors provide chemically plausible hard negatives. These negatives are deliberately more informative than random non-relevant documents: they ask whether the retriever can distinguish the right chemical evidence from evidence that is close enough to be tempting.

The resulting benchmark labels each retrieved document by its role in the workflow: same-language relevant evidence, cross-language relevant evidence, unrelated distraction, or chemically confusable hard negative. The next section uses these labels to evaluate multilingual embedding models beyond aggregate recall.

5 Evaluation Setup

TODO: Write this section from the paper plan.

6 Results

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Limitations

TODO: Write this section from the paper plan.

7 Conclusion

TODO: Write this section from the paper plan.

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A Appendix

TODO: Optional appendix material from the paper plan.